Day 126 | Unit 09 | Teacher plan

Display, Describe, and Summarize Data

Use Frequency Tables and Histograms

Standards: 6.SP.B.4

Learning target: I can construct and interpret a histogram with equal-width bins.

enVision Topic 8: Display, Describe, and Summarize Data | Spectrum companion Chapter 7

0–5 minutes / Launch

Read the target. Ask for an estimate or a sketch before calculating. Connect the new idea to the previous lesson.

5–17 / Model

Use Worked Example 1, then Example 2. Ask students to name the next step and explain a choice. Group numerical values into adjacent, nonoverlapping intervals. Count each observation in exactly one interval, draw touching bars, and label the numerical axis and the frequency axis.

17–42 / Practice

Assign questions 1–10 in order. Use 1–3 for foundations, 4–7 for application, and 8–10 for state-format response and reasoning. The example button provides a separate question with a worked method.

42–50 / Discuss and check

Compare representations. Review a selected answer and a plausible error. Use question 10 as an exit discussion; ask students to check units, scale, or substitution.

Support and extension

Offer number lines, fraction strips, grid paper, or a table template as appropriate. Read multi-part directions one part at a time. For extension, ask for another representation or a counterexample.

Companion and state alignment

Use the matching skill in the edition you own. No unverified publisher page numbers are assigned. AASA references below identify related public sample items by standard and format; all course questions are original.

AASA sample item references: 13

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01 / Learn

Group numerical values into adjacent, nonoverlapping intervals. Count each observation in exactly one interval, draw touching bars, and label the numerical axis and the frequency axis.

Worked example 1

Daily reading times in minutes: 13, 16, 14, 15, 0, 14. Draw a histogram using bins 0–4, 5–9, 10–14, and 15–19. Label axes.

1. Group numerical values into adjacent, nonoverlapping intervals.

2. Count each observation in exactly one interval, draw touching bars, and label the numerical axis and the frequency axis.

3. Count every value once.

4. Bars touch and have equal width

5. horizontal axis is minutes, vertical axis is frequency.

6. State the result in the form requested: Frequencies 1, 0, 3, 2.

Answer: Frequencies 1, 0, 3, 2

Worked example 2

Reading-time frequencies (minutes) are 0–4: 3, 5–9: 4, 10–14: 2, 15–19: 2. Part A: draw a histogram. Part B: enter the observation count and name every highest-frequency interval.

1. Group numerical values into adjacent, nonoverlapping intervals.

2. Use four equal-width touching bars with heights 3, 4, 2, 2.

3. Frequencies sum to 11.

4. Include every interval with frequency 4.

Answer: 11 observations; highest intervals 5–9

Day 126 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Daily reading times in minutes: 5, 18, 18, 13, 9, 17. Draw a histogram using bins 0–4, 5–9, 10–14, and 15–19. Label axes.

02 / Constructed response

Daily reading times in minutes: 18, 5, 14, 0, 19, 18. Draw a histogram using bins 0–4, 5–9, 10–14, and 15–19. Label axes.

03 / Constructed response

Daily reading times in minutes: 18, 17, 14, 6, 13, 1. Draw a histogram using bins 0–4, 5–9, 10–14, and 15–19. Label axes.

04 / Constructed response

Daily reading times in minutes: 0, 2, 5, 7, 2, 4, 15, 16, 9, 5. Draw a histogram using bins 0–4, 5–9, 10–14, and 15–19. Label axes.

05 / Constructed response

Daily reading times in minutes: 14, 5, 9, 17, 7, 0, 15, 18, 8, 16. Draw a histogram using bins 0–4, 5–9, 10–14, and 15–19. Label axes.

06 / Constructed response

Daily reading times in minutes: 14, 6, 10, 0, 1, 18, 6, 2, 11, 1. Draw a histogram using bins 0–4, 5–9, 10–14, and 15–19. Label axes.

07 / Constructed response

Daily reading times in minutes: 7, 9, 13, 2, 15, 12, 16, 0, 17, 8. Draw a histogram using bins 0–4, 5–9, 10–14, and 15–19. Label axes. Which interval has the greatest frequency? Include ties.

08 / Data display and explanation

Reading-time frequencies (minutes) are 0–4: 2, 5–9: 3, 10–14: 5, 15–19: 5. Part A: draw a histogram. Part B: enter the observation count and name every highest-frequency interval.

09 / Data display and explanation

Reading-time frequencies (minutes) are 0–4: 4, 5–9: 2, 10–14: 1, 15–19: 2. Part A: draw a histogram. Part B: enter the observation count and name every highest-frequency interval.

10 / Constructed response

Daily reading times in minutes: 16, 13, 0, 7, 10, 1, 2, 1, 4, 18. Draw a histogram using bins 0–4, 5–9, 10–14, and 15–19. Label axes. Which interval has the greatest frequency? Include ties.

Day 126 | Unit 09 | Answer key

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1. Frequencies 0, 2, 1, 3

Count every value once. Bars touch and have equal width; horizontal axis is minutes, vertical axis is frequency.

6.SP.B.4 | Constructed response

2. Frequencies 1, 1, 1, 3

Count every value once. Bars touch and have equal width; horizontal axis is minutes, vertical axis is frequency.

6.SP.B.4 | Constructed response

3. Frequencies 1, 1, 2, 2

Count every value once. Bars touch and have equal width; horizontal axis is minutes, vertical axis is frequency.

6.SP.B.4 | Constructed response

4. Frequencies 4, 4, 0, 2

Count every value once. Bars touch and have equal width; horizontal axis is minutes, vertical axis is frequency.

6.SP.B.4 | Constructed response

5. Frequencies 1, 4, 1, 4

Count every value once. Bars touch and have equal width; horizontal axis is minutes, vertical axis is frequency.

6.SP.B.4 | Constructed response

6. Frequencies 4, 2, 3, 1

Count every value once. Bars touch and have equal width; horizontal axis is minutes, vertical axis is frequency.

6.SP.B.4 | Constructed response

7. Frequencies 2, 3, 2, 3; greatest: 5–9, 15–19

Count every value once. Bars touch and have equal width; horizontal axis is minutes, vertical axis is frequency.

6.SP.B.4 | Constructed response

8. 15 observations; highest intervals 10–14, 15–19

Use four equal-width touching bars with heights 2, 3, 5, 5. Frequencies sum to 15. Include every interval with frequency 5.

6.SP.B.4 | Data display and explanation

9. 9 observations; highest intervals 0–4

Use four equal-width touching bars with heights 4, 2, 1, 2. Frequencies sum to 9. Include every interval with frequency 4.

6.SP.B.4 | Data display and explanation

10. Frequencies 5, 1, 2, 2; greatest: 0–4

Count every value once. Bars touch and have equal width; horizontal axis is minutes, vertical axis is frequency.

6.SP.B.4 | Constructed response